CMSC5743 Lab 02

Convolution Acceleration

1 Assignments

Q1 – Strassen Matrix Multiplication Implement the matrix multiplication using Strassen Algorithm and compare the speed with original `matmul()` in Lab 01. The shape of matrix A is $I \times K$ and the shape of matrix B is $K \times J$. The matrix size setting remains the same as Lab 01, the value of I, K, J will be fixed at 256, 512 or 1024.

Q2 – Winograd Convolution Implement a C++ version from scratch based on Winograd algorithm $F(2 \times 2, 3 \times 3)$ and compare the speed with your original `im2col` implementation in Lab 01. Please provide analysis on whether or not your implementation improves the speed performance and why. The convolution kernel and input size remain the same as Lab 01:

- batch: 1
- height_feature: 56
- width_feature: 56
- in_channels: 3
- out_channels: 64
- kernel_size: 3
- stride: 1
- padding: 0

2 Submission Requirements

2.1 Directory Structure

Submit a single `.zip` file named `Lab2_<StudentID>.zip` with the following structure:

```
Lab2_<StudentID>/
  |- report.pdf
  |- code/
  |    |- strassen_impl.cpp        (Q1 implementation)
  |    |- winograd_impl.cpp        (Q2 implementation)
  |    |- Makefile
  |- README.txt   (build & run instructions, optional notes)
```

2.2 Source Code Requirements

1. **Use the provided template files.** Implement the functions marked with `TODO` in `strassen_impl.cpp` and `winograd_impl.cpp`. Do **not** modify the function signatures listed below.

2. **Q1 required interfaces** (in `strassen_impl.cpp`):
   - `void matrix_add(float** A, float** B, float** C, int n)`
   - `void matrix_sub(float** A, float** B, float** C, int n)`
   - `void strassen(float** A, float** B, float** C, int n)`
   - `void strassen_multiply(const float* A, int I, int K, const float* B, int K2, int J, float* C)`

3. **Q2 required interfaces** (in `winograd_impl.cpp`):
   - `void kernel_transform(const float* g, float* U)`
   - `void input_transform(const float* d, int tile_stride, float* V)`
   - `void output_transform(const float* M, float* Y, int out_stride)`
   - `void winograd_conv2d(const float* input, int C_in, int H, int W, const float* kernel, int C_out, int K_size, float* output, int stride, int pad)`

4. **No external libraries.** Your code must compile with only the C++ standard library. Do not use Armadillo, Eigen, BLAS, or similar libraries.

5. **Compilation.** Your code must compile cleanly with:

```
g++ -std=c++17 -O3 -Wall -o strassen strassen_impl.cpp
g++ -std=c++17 -O3 -Wall -o winograd winograd_impl.cpp
```

6. You may add **helper functions**, but do not remove or rename the required interfaces.

2.3 Report Requirements

The report (`report.pdf`, max 4 pages) should include:

1. **Q1 Analysis:**
   - Runtime comparison table: Naive vs. Strassen for $n \in \{256, 512, 1024\}$.
   - Speedup analysis. Discuss whether Strassen is faster in practice and why (e.g., recursion overhead, cache effects, crossover point).

2. **Q2 Analysis:**
   - Runtime comparison table: Direct convolution (or `im2col` from Lab 01) vs. Winograd.
   - Analysis of whether Winograd improves performance and **why** (e.g., FLOPs reduction, memory access pattern, transform overhead).

3. **Correctness:** Show that your implementations produce correct results (e.g., max error compared to reference).

Useful Materials

- Strassen Algorithm
- Fast Algorithms for Convolutional Neural Networks (Lavin & Gray, 2015)